

Marking Scheme

2026 · Paper Paper 1 · 105 marks

General Rules

- M / A / PROOF mark types
- **f.t.** : M marks always follow through. A marks follow through ONLY when isFollowThrough = true.
- Deduction cap : A(1)=1 · A(2)=1 · B=0 (u, pp)
- No deduction in any step scoring 0.
- A correct answer merits all marks unless a method is specified.
- Different notation is NOT penalized. · Benefit of doubt
- All fractional answers must be simplified.

Section A(1) (35 marks)

1. Change of Subject [B-]

a (3 marks)

M1 $4x - 3 = c(x + 2c)$

1M for cross-multiplication / 交叉相乘

M1 $4x - cx = 2c^2 + 3 \Rightarrow x(4 - c) = 2c^2 + 3$

1M for grouping x terms / 合併 x 項

A1 $x = \frac{2c^2 + 3}{4 - c}$

2. Laws of Indices [B-]

a (3 marks)

M1 $(m^{-2}n^5)^{-3} = m^6n^{-15}$

1M for power of a power

M1 $(mn^{-1})^4 = m^4n^{-4}; \frac{m^6n^{-15}}{m^4n^{-4}} = m^2n^{-11}$

A1 $\frac{m^2}{n^{11}}$

3. Estimation and Error [B]

a (3 marks)

M1 ***Weight of one packet lies in*** [470, 490] g

1M for using 480 +/- 10

M1 ***Least possible total*** = $150 \times 470 = 70500 \text{ g} = 70.5 \text{ kg}$

A1 ***70.4 kg (tonearest 0.1 kg) means total < 70.45 < 70.5, so the claim is NOT correct***

1A must state NOT correct with reason / 須說明不正確並附理由

4. Compound Inequalities [B]

a (3 marks)

M1 $x + 5 > 2x - 4 \Rightarrow x < 9$

M1 $4x + 15 \geq x \Rightarrow 3x \geq -15 \Rightarrow x \geq -5$

A1 ***Taking the intersection :*** $-5 \leq x < 9$

b (1 mark)

A1 ***Negative integers are*** $-5, -4, -3, -2, -1$; ***there are*** 5 ***ft.***

1A f.t. from (a)

5. Percentages [B]

a (4 marks)

M1 ***Let the number of Chinese books be*** c ; ***number of English books*** = $1.2c$

M1 $c = 1.2(1.2c - 66)$

1M for forming the equation / 列方程

M1 $c = 1.44c - 79.2 \Rightarrow 0.44c = 79.2$

A1 $c = 180$

6. Ratio and Proportion [B]

a (2 marks)

M1 $3a = 2b \Rightarrow a : b = 2 : 3; \text{ let } a = 2t, b = 3t$

A1 $\frac{9t - c}{2t} = \frac{3}{2} \Rightarrow c = 6t; a : b : c = 2 : 3 : 6$

b (2 marks)

M1 $\frac{4(2t) + 3(6t)}{5(3t) - 2(6t)} = \frac{26t}{3t}$ ft.

A1 $= \frac{26}{3}$ ft.
1 A f.t. from (a)

7. Properties of Circles [B+]

a (4 marks)

M1 $\angle ACB = \angle ADB = 36^\circ$ (angles in the same segment)

A1 $\angle BEC = 180^\circ - 100^\circ = 80^\circ; \text{ in } \triangle BEC, \angle DBC = 180^\circ - 80^\circ - 36^\circ = 64^\circ$

M1 $\angle ABC = 90^\circ$ (in semi-circle); $\angle BAC = 90^\circ - 36^\circ = 54^\circ$

A1 $\angle CAD = \angle DBC = 64^\circ$ (angles in the same segment); $\angle BAD = 54^\circ + 64^\circ = 118^\circ$ ft.

8. Similar Triangles [B+]

a (2 marks)

M1 $\angle PTR = \angle QTS$ (vert. opp. angles); $\angle RPT = \angle SQT$ (alt. angles, $PR \parallel SQ$)

PROOF $\therefore \triangle PRT \sim \triangle QST$ (AAA)

b (3 marks)

M1 $\frac{PT}{QT} = \frac{PR}{QS} = \frac{9}{15} = \frac{3}{5}$ and $PT + QT = 40 \Rightarrow PT = 15, QT = 25$

M1 $\frac{RT}{ST} = \frac{3}{5} \Rightarrow ST = \frac{5}{3}(12) = 20 \text{ cm}$

A1 $QS^2 + ST^2 = 15^2 + 20^2 = 625 = 25^2 = QT^2$, so $\triangle QST$ is right-angled at S (converse of Pyth. thm)
ft.

9. Stem-and-Leaf Diagram and Probability [B]

a (3 marks)

M1 $\text{Range} = (30 + a) - 12 = 26 \Rightarrow a = 8$

A1 $\text{Mean} = \frac{432}{18} = 24 \text{ hours}$
u-1 if unit missing

A1 $\text{Mode} = 23 \text{ hours}$

b (2 marks)

M1 $\text{Number of workers with working hours} > 23 : 25, 26, 29, 30, 32, 34, 35, 38 \Rightarrow 8$ ft.

A1 $P = \frac{8}{18} = \frac{4}{9}$ ft.

Section A(2) (35 marks)

10. Locus and Circles [A-]

a (1 mark)

A1 Γ is the perpendicular bisector of AB

b (2 marks)

M1 $AB \perp \Gamma; \text{ slope of } \Gamma = -2 \Rightarrow \text{ slope of } AB = \frac{1}{2}$

A1 $y - 2 = \frac{1}{2}(x + 3) \Rightarrow x - 2y + 7 = 0$

c (3 marks)

M1 $\text{Mid-point of } AB = \text{intersection of } AB \text{ and } \Gamma : \text{ solving gives } (1, 4)$

M1 $B = (5, 6); \text{ radius} = \sqrt{(1+3)^2 + (4-2)^2} = \sqrt{20}$ ft.

A1 $(x-1)^2 + (y-4)^2 = 20$, i. e. $x^2 + y^2 - 2x - 8y - 3 = 0$ ft.

11. Frequency Distribution and Dispersion [A-]

a (5 marks)

M1 *Total number of students = 30; cumulative frequencies 2, 9, 21, 28, 30*

A1 *Median = $\frac{15\text{th} + 16\text{th}}{2} = 2$*

M1 *$Q_1 = 8\text{th datum} = 1, Q_3 = 23\text{rd datum} = 3$*

A1 *IQR = $3 - 1 = 2$*

A1 *Variance = $\frac{2(2)^2 + 7(1)^2 + 12(0)^2 + 7(1)^2 + 2(2)^2}{30} = \frac{30}{30} = 1$*

b (2 marks)

M1 *Mean = 2; unchanged mean $\Rightarrow$ the two students own a total of 4 calculators, i. e. (0, 4), (1, 3) or (2, 2)*

A1 *In each case at least one student owning 0 and at least one owning 4 remain, so the range remains $4 - 0 = 4$: NO change*

12. Partial Variation and Circles [A-]

a (3 marks)

M1 *Let $f(x) = h + kx^2$, where $k \neq 0$*

M1 *$h + 4k = 9, h + 25k = 30 \Rightarrow 21k = 21$*

A1 *$k = 1, h = 5 \Rightarrow f(x) = 5 + x^2$*

b (4 marks)

M1 *$U = (0, 5), V = (3, 14), W = (0, 14)$ **ft.***

M1 *$UW \perp WV \Rightarrow \angle UWW = 90^\circ \Rightarrow UV$ is a diameter of C
IM for recognising angle in semi-circle / 認出半圓上的圓周角*

M1 *$UV = \sqrt{3^2 + 9^2} = \sqrt{90} = 3\sqrt{10}$*

A1 *Circumference = $3\sqrt{10}\pi$ **ft.***

13. Polynomials and Rational Roots [A]

a (3 marks)

M1 *Let the quotient be $px + r$; then $h(x) = g(x)(px + r) + (px + r) = (px + r)[g(x) + 1]$
IM for common factor / 提取公因式*

M1 *Comparing coefficients of x^4 : $p = 2; g(x) + 1 = x^3 + 2x^2 - 5x + 4$; coefficient of x^2 : $-10 + 2r = -8$*

A1 *$r = 1$; the quotient is $2x + 1$*

b (4 marks)

M1 *$h(x) = (2x + 1)(x^3 + 2x^2 - 5x + 4)$ **ft.***

A1 *$2x + 1 = 0$ gives the rational root $x = -\frac{1}{2}$*

M1 *By the rational root theorem any rational root of $x^3 + 2x^2 - 5x + 4 = 0$ must be $\pm 1, \pm 2$ or ± 4 ; testing shows none is a root, so the cubic has NO rational roots*

A1 *$h(x) = 0$ has exactly 1 rational root **ft.***

14. Mensuration [A]

a (3 marks)

M1 *$\pi r l = 240\pi \Rightarrow 12l = 240 \Rightarrow l = 20$ cm*

M1 *$h = \sqrt{l^2 - r^2} = \sqrt{20^2 - 12^2} = \sqrt{256}$*

A1 *$h = 16$ cm
u-1 if unit cm missing*

b (3 marks)

M1 *Curved SA of original cone = $4 \times$ curved SA of $X \Rightarrow$ ratio of areas = $1 : 4$*

M1 *Ratio of lengths = $1 : 2 \Rightarrow$ ratio of volumes = $1 : 8$*

A1 *Volume of cone = $\frac{1}{3}\pi(12)^2(16) = 768\pi$; volume of $Y = \frac{7}{8}(768\pi) = 672\pi$ cm³ **ft.***

c (2 marks)

M1 *$\pi(4)^2 h = 672\pi$ **ft.***

A1 *$h = 42$ cm **ft.**
u-1 if unit cm missing*

Section B (35 marks)

15. Probability [A-]

a (2 marks)

M1 $P = \frac{C_2^6}{C_2^9} = \frac{15}{36}$

A1 $= \frac{5}{12}$

b (2 marks)

M1 $P = \frac{5}{12}(1) + \frac{1}{2} \cdot \frac{C_3^8}{C_3^9} + \frac{1}{12} \cdot \frac{C_3^7}{C_3^9}$

1M for considering all three transfer cases / 考慮三種轉移情況

A1 $= \frac{5}{12} + \frac{1}{3} + \frac{5}{144} = \frac{113}{144}$ ft.

16. Quadratic Equations and Circles [A]

a (2 marks)

M1 **Sum of roots** : $q + 3q = 4q = -a \Rightarrow q = -\frac{a}{4}$; **product** : $3q^2 = b$

PROOF $3\left(-\frac{a}{4}\right)^2 = b \Rightarrow \frac{3a^2}{16} = b \Rightarrow 3a^2 = 16b$

b (3 marks)

M1 **Substituting** $y = mx : (1 + m^2)x^2 - (6 + 2m)x + 6 = 0$

M1 $OQ : QR = 1 : 2 \Rightarrow \text{the roots are in the ratio } 1 : 3; \text{ by (a) } 3A^2 = 16B \text{ with } A = -\frac{6 + 2m}{1 + m^2}, B = \frac{6}{1 + m^2}$

A1 $3(6 + 2m)^2 = 96(1 + m^2) \Rightarrow 7m^2 - 6m - 1 = 0 \Rightarrow m = 1 \text{ or } m = -\frac{1}{7}$

17. Trigonometry in 3-D [A]

a (2 marks)

M1 **By the sine formula** : $\frac{\sin \angle ACB}{7} = \frac{\sin 60^\circ}{9}$

A1 $\angle ACB \approx 42.3^\circ$
1A r.t. 42.3 degrees
r.t. 42.3

b (4 marks)

M1 $PA = PB = PC \Rightarrow \text{the foot O of the perpendicular from P is the circumcentre of } \triangle ABC$

M1 **Circumradius** $R = \frac{BC}{2 \sin \angle BAC} = \frac{9}{2 \sin 60^\circ} \approx 5.196$; $PO = R \tan 40^\circ \approx 4.360$

M1 **Let M be the mid-point of BC**; $OM = R \cos \angle BAC \approx 2.598$ and the required angle is $\angle PMO$

A1 $\tan \angle PMO = \frac{4.360}{2.598} \Rightarrow \angle PMO \approx 59.2^\circ > 55^\circ$, so YES it exceeds 55° ft.
r.t. 59.2

18. Geometric Sequences and Logarithms [A+]

a (3 marks)

M1 $\log_b a = \frac{\log a}{\log b}$ and $\log_a b = \frac{\log b}{\log a}$

M1 $1 < a < b \Rightarrow \log a \neq 0$ and $\log b \neq 0$

A1 $\log_b a = \frac{1}{\log_a b}$

b (5 marks)

M1 $a, c, b \text{ geometric} \Rightarrow c^2 = ab \Rightarrow \log c = \frac{\log a + \log b}{2}$

M1 **Let** $t = \log_a b$; then $\log_c b = \frac{2t}{1+t}$ and $\log_b a = \frac{1}{t}$

M1 **Arithmetic** $\Rightarrow 2 \left(\frac{2t}{1+t} \right) = \frac{1}{t} + t$

M1 $4t^2 = (1+t^2)(1+t) \Rightarrow t^3 - 3t^2 + t + 1 = 0 \Rightarrow (t-1)(t^2 - 2t - 1) = 0$

A1 $1 < a < b \Rightarrow t > 1 \Rightarrow t = 1 + \sqrt{2}$; **common difference** $= \frac{2t}{1+t} - \frac{1}{t} = \sqrt{2} - (\sqrt{2} - 1) = 1$

19. Coordinate Geometry [A+]

a (2 marks)

M1 OB is a median $\Rightarrow B$ is the mid-point of AC

A1 $C = (2(3t) - 30, 2t - 0) = (6t - 30, 2t)$

b (4 marks)

M1 $D = (3t, 0); \tan \angle AOB = \frac{t}{3t} = \frac{1}{3}$

M1 $\tan \angle ABD = \frac{AD}{BD} = \frac{30 - 3t}{t}$

M1 $\tan 2\angle AOB = \frac{2(1/3)}{1 - (1/3)^2} = \frac{3}{4}$

PROOF $\frac{30 - 3t}{t} = \frac{3}{4} \Rightarrow 120 - 12t = 3t \Rightarrow t = 8$

c (3 marks)

M1 $t = 8 \Rightarrow C = (18, 16); G$ lies on the perpendicular bisector of OA , i.e. $x = 15$

M1 $GO^2 = GC^2: 15^2 + y^2 = (15 - 18)^2 + (y - 16)^2$

A1 $225 = 9 + 256 - 32y \Rightarrow y = \frac{5}{4}; G = \left(15, \frac{5}{4}\right)$

d (3 marks)

M1 $\text{Slope of } OG = \frac{5/4}{15} = \frac{1}{12}$ (ft.)

M1 $\text{Slope of } OB = \frac{8}{24} = \frac{1}{3}$

A1 $\frac{1}{12} \neq \frac{1}{3}$, so O, G and B are NOT collinear (ft.)
